Project Planck Submission — Paralog Redundancy / Curative Threshold

Prompt (submit this block)

K1, K2, K3 are three paralogous kinase proteins co-expressed in six hypothetical tumor cell lines (L1-L6). siK1, siK2, siK3 are validated siRNAs; each silences expression of only its correspondingly numbered protein (confirmed by immunoblot, equivalent knockdown efficacy across all three, no detectable reduction of either other paralog).

Table: proliferation (% of non-targeting control) / death-marker (normalized to a cytotoxic positive control = 1.00), by treatment and cell line.

Treatment	L1	L2	L3	L4	L5	L6
siK1	100/0.10	99/0.11	100/0.09	101/0.12	99/0.10	100/0.11
siK2	99/0.11	100/0.10	101/0.10	100/0.09	100/0.12	99/0.10
siK3	100/0.10	100/0.12	99/0.11	99/0.10	101/0.11	100/0.09
siK1+siK2	99/0.12	100/0.10	100/0.11	101/0.11	99/0.09	100/0.10
siK2+siK3	100/0.11	99/0.10	100/0.10	100/0.12	100/0.10	101/0.11
siK1+siK3	**98/0.12**	**45/0.22**	**52/0.18**	**95/0.10**	**101/0.15**	**40/0.25**
siK1+siK2+siK3	**8/0.82**	**5/0.91**	**6/0.88**	**9/0.79**	**7/0.85**	**4/0.93**

Define a curative response in a given line as death-marker exceeding 0.60 in that line.

Using only the table above, in how many of the six cell lines does the siK1+siK3 treatment cross the curative threshold? Report as a single integer.

Character count: 1,324 / 2,000 limit (highlighted rows for illustration only — remove highlighting before actual submission so the model isn't cued toward which rows matter)

Answer Format

Integer, 0–6.

Final Answer

0

Golden Response

Step 1: Define the notation.
K1, K2, K3 are the three paralogous proteins. siK1, siK2, siK3 are the siRNAs that each silence only their correspondingly numbered protein, with no cross-silencing of the other two (stated explicitly). A treatment name (e.g. "siK1+siK3") identifies which siRNAs were applied; it does not itself state which protein remains unsilenced — that must be derived: silencing K1 and K3 leaves K2 as the only protein of the three not silenced.

Step 2: Define what "curative" means.
A curative response in a given line is defined as death-marker exceeding 0.60 in that line specifically. Proliferation percentage is a separate column and does not by itself indicate a curative response, however low it drops.

Step 3: Locate the correct table row.
The question concerns the siK1+siK3 row. Its six death-marker values (second number in each cell) are: L1 0.12, L2 0.22, L3 0.18, L4 0.10, L5 0.15, L6 0.25.

Step 4: Apply the threshold to each value.
L1: 0.12 < 0.60 — no
L2: 0.22 < 0.60 — no
L3: 0.18 < 0.60 — no
L4: 0.10 < 0.60 — no
L5: 0.15 < 0.60 — no
L6: 0.25 < 0.60 — no

None of the six death-marker values in the siK1+siK3 row exceed 0.60, even though proliferation in that same row is substantially reduced in three lines (L2 45%, L3 52%, L6 40%). A response that reads the proliferation numbers instead of the death-marker numbers, or that treats "proliferation reduced in 3 lines" as equivalent to "curative in 3 lines," will answer 3 instead of 0 — the specific reasoning error this question is built to catch.

Step 5: Final synthesis.
Zero of the six cell lines cross the curative threshold under the siK1+siK3 treatment. Leaving even one of the three paralogous proteins unsilenced (here, K2) is never sufficient, on its own, to allow a curative response to occur — regardless of how much that treatment suppresses proliferation in a given line. Only the siK1+siK2+siK3 row crosses the threshold, and does so in all six lines.

Final Answer: 0

Notes on Originality / Design (not part of submission)

- K1/K2/K3, siK1/siK2/siK3, and the six-line panel are entirely fictional.
- The five "no effect" treatment rows are given real per-line numbers (small, plausible noise around baseline) rather than being summarized in prose — the model must scan the full table itself to determine which rows vary meaningfully, not be told in advance which two rows matter.
- Two reasoning steps are required: (1) derive that K2 is the unsilenced protein under siK1+siK3 (never stated directly), and (2) read the death-marker column, not the proliferation column, against the stated threshold. Skipping either produces a plausible-looking but wrong answer (3, from miscounting proliferation-reduced lines as curative).